

Multiple Choice

1. **Answer:** C

Options: A) Low motor unit strength; B) Low glycogen storage capacity; C) High fatigue resistance; D) Low oxidative capacity; E) Low glycolytic capacity

Note: Correct option: C - High fatigue resistance

2. **Answer:** D

Options: A) Increases in life expectancy at a specified age; B) The increase in the rate of technological advancements; C) The decrease in childhood mortality rates; D) The steadily increasing birth rate; E) The Baby Boom generation

Note: Correct option: D - The steadily increasing birth rate

3. **Answer:** D

Options: A) lipogenesis.; B) lipolysis.; C) lipo-genesis.; D) gluconeogenesis.; E) proteolysis.

Note: Correct option: D - gluconeogenesis.

4. **Answer:** C

Options: A) 10-15%; B) >20%; C) <10%; D) >15%; E) <5%

Note: Correct option: C - <10%

5. **Answer:** A

Options: A) lncRNA; B) piRNA; C) rRNA; D) snRNA; E) tRNA

Note: Correct option: A - lncRNA

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '46,XX,der(14;21)(q 10;q 10)pat+21'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Human'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Vitamin A. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The third and fourth pharyngeal arches. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key